

Hao Zheng

Recent M.S. Graduate | Tsinghua University

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ORCID | Google Scholar | Mobility Lab | Website

CORE METHOD AREAS

Machine Learning: Time Series Forecasting | Graph Neural Networks | Sequence-to-Sequence Models | Transfer Learning | Knowledge-Guided Learning | Multi-Task Learning | Discrete Choice Modeling

Optimization & Operations Research: Mixed-Integer Programming | Bilevel Optimization | Frank-Wolfe | Column Generation | Branch-and-Price | VRP/VRPTW | ALNS | Tabu Search

Transportation & Infrastructure Modeling: Space-Time Network Modeling | Traffic Assignment | GMNS | Microscopic Traffic Simulation | Travel Demand Modeling | Spatial Network Analysis | Pavement Performance Prediction | Pavement Maintenance Optimization

EDUCATION

- **Tsinghua University | M.S. in Transportation Engineering** 09/2022 – 06/2025
Department of Civil Engineering, Beijing, China GPA: **3.86/4.00**
Thesis: A Comprehensive Full-Process Planning Model for Pavement Performance Forecasting and Maintenance Strategy Implementation under Missing Data Conditions.
- **Beijing Jiaotong University | B.Eng. in Traffic and Transportation** 09/2018 – 06/2022
School of Traffic and Transportation, Beijing, China GPA: **3.80/4.00**; Weighted Average: **88.3/100**
Concentration: Railway Transportation.
Thesis: Customized Bus Planning Method Responding to Spatiotemporal Concentrated Demand in Super-Peak.

CURRENT RESEARCH & PROFESSIONAL ACTIVITIES

- **Open-Source Project Developer**, Mobility Computation Lab 2025 – Present
Developed under the guidance of Prof. Xuesong Zhou 🌐 📄
- **Research Assistant**, Beijing Jiaotong University 07/2025 – Present
Modernization Laboratory for Intelligent Rail Transit Organization and Management
- **Associate Member**, PIARC Technical Committee 1.1 (World Road Association) 2026 – Present

PUBLICATIONS

- **Hao Zheng**, Han Zheng*, George Giannopoulos. KNOWCAST: Domain-knowledge-integrated deep learning for metro OD forecasting and path flow estimation. DOI:10.1016/j.trc.2026.105710 🌐 *Transportation Research Part C: Emerging Technologies*, 189, 105710. 2026
Method focus: Graph Neural Networks | Sequence-to-Sequence Models | Knowledge-Guided Learning
- **Hao Zheng**, Han Zheng*, George Giannopoulos. Integrated Routing of Trucks and Robots for Multi-Modal Urban Express Logistics. 🌐 *Presented at the TRB 105th Annual Meeting, 2026; Transportation Research Record, Accepted for publication.* 2026
Method focus: Space-Time Network Modeling | Branch-and-Price | Column Generation | C#/Gurobi
- **Hao Zheng**, Jiantao Li, Yian Zhao, Xi Zhu, Runhua Guo*. An operational scheduling framework for pavement maintenance activities: from rule-based task generation to branch-and-price-based route optimisation. DOI:10.1080/10298436.2026.2708283 🌐 *International Journal of Pavement Engineering, Article 2708283.* 2026
Method focus: VRPTW | Road Network Modeling | Branch-and-Price | Column Generation
- Jiantao Li, Yian Zhao, **Hao Zheng**, Runhua Guo. Vehicle Vibration Modelling via Road Profile Reconstruction and Simulation: Coupled Analysis of IRI, Speed, and Ride Comfort for Driving Assistance. DOI:10.1080/14680629.2025.2574300. *Road Materials and Pavement Design.* 2025
Method focus: Road Profile Reconstruction | Vehicle Dynamics Simulation | IRI-Based Modeling
- Yuyan Tan, Yibo Li, Ruxin Wang, Xiwei Mi, Yaxuan Li, **Hao Zheng**, Yu Ke, Yan Wang. Improving Synchronization in High-Speed Railway and Air Intermodality: Integrated Train Timetable Rescheduling and Passenger Flow Forecasting. *IEEE Transactions on Intelligent Transportation Systems*, 23(3), 2651–2667. DOI:10.1109/TITS.2021.3137410. 2022
Method focus: Timetable Rescheduling | Passenger Flow Forecasting | Multi-Objective Optimization
- **Hao Zheng**, Xingchen Zhang, Junhua Chen. Study on Customized Shuttle Transit Mode Responding to Spatiotemporal Inhomogeneous Demand in Super-Peak. *Information*, 12(10), 429. DOI:10.3390/info12100429 🌐 2021
Method focus: Pickup-and-Delivery with Time Windows | ALNS | Tabu Search
- **Hao Zheng**, Yun Wang*, Xuedong Yan, Yunwei Li. Dynamic Planning Model and Solution Algorithm for the Demand-responsive Customized Bus Route. *Proceedings of the World Transport Convention / World Transport Engineering and Technology Forum (WTC2021)*, People's Communications Press, 2021. ISBN: 9787114173554. 🌐 2021
Method focus: Two-Stage Planning | Dynamic Insertion | Route Re-Optimization

MANUSCRIPTS UNDER REVIEW AND IN PREPARATION

- **Hao Zheng**, Runhua Guo*, Xi Zhu, Jiantao Li, Long Wen, Weining Zhou. Public-Data-Assisted Staged Transfer Learning for Local Pavement Condition Prediction across Shared and Local Indicators.  *Submitted to International Journal of Pavement Engineering. 2026*
Method focus: [Transfer Learning](#) [Multi-Task Learning](#) [Sequence-to-Sequence Models](#)
- **Hao Zheng**, Han Zheng*, George Giannopoulos, et al. Heat-related road conditions and cycling network continuity across European cities. *Manuscript in preparation for submission to Nature Cities. 2026*
Method focus: [Spatial Network Analysis](#) [Network Robustness](#) [Accessibility Modeling](#) [Remote Sensing](#)
- **Hao Zheng**, Han Zheng*, George A. Giannopoulos, et al. DISCERN: Discrete Choice with Embedded Representation Networks. *Working paper, 2026. 2026*
Method focus: [Discrete Choice Models](#) [Representation Learning](#) [Gradient-Based Estimation](#)

SELECTED RESEARCH EXPERIENCE

- **Mobility Computation Lab**   2025 – Present
Open-source research project with Prof. Xuesong Zhou
 - Built a GMNS-based platform connecting city networks, travel demand, supported observations, and reproducible network computation across documented Boston, Sioux Falls, and Hong Kong cases.
 - Implemented Frank–Wolfe, Algorithm B, finite-path/compressed assignment, and finite time-expanded decomposition workflows with reusable adapters and independent feasibility and original-space checks.
- **Computational Benchmarking of Scalable Traffic Assignment on Open Transportation Networks** 2025 – Present
Research Collaborator with Prof. Xuesong Zhou
 - Built an experimental pipeline converting transportation benchmark networks into unified link–demand formats and generating path sets for path-based traffic assignment.
 - Evaluated path-compression strategies across compression ratios and demand levels, comparing solver time, objective gaps, and link-flow fit against baseline solutions.
- **Open Mobility Data Visibility: A Global Audit of Public Urban Mobility Data** 2026 – Present
Ongoing Research Project
 - Developed a reproducible global audit covering 11,422 urban centres and integrating public transit directories, GTFS content, OpenStreetMap, GBFS, official/API entry points, and multilingual evidence.
 - Distinguished source discoverability, downloadable-data availability, parseability, spatially verifiable service visibility, and review-only evidence through versioned packages and city-level audit records.
- **Heat-related Road Conditions and Cycling Network Continuity across European Cities** 2026 – Present
Ongoing Research Project
 - Built a 23-city comparative framework linking low-stress cycling networks with land-surface temperature, tree-cover deficit, built-up context, population, and destinations.
 - Evaluated connectivity, accessibility, route availability, and detour after treating the highest-scoring 5%, 10%, and 20% of roads as unavailable, with comparisons against random road sets matched by count or approximate total length.
- **Bilevel Optimization Framework for Tolling Strategy in Autonomous Networks**  06/2025 – 10/2025
MIT-UF-NEU Summer Research Camp | Program PI: Prof. Jinhua Zhao, MIT; Mentor: Dr. Xuan Jiang
 - Designed an end-to-end bilevel optimization workflow for tolling strategy design, modeling interactions between infrastructure operators and autonomous vehicle fleets.
 - Implemented the Python-based upper-level optimization loop and re-engineered a C++/CUDA microscopic traffic simulator to support toll-responsive agent behavior.
- **Intelligent Road Asset Management: Prediction and Decision-Support Models** 2023 – 2025
Digital Transportation Management Platform | Supervisor: Prof. Runhua Guo
 - Developed core prediction and maintenance decision modules for a digital road asset management platform, integrating pavement condition forecasting with maintenance strategy optimization.
 - Built GRU/attention-based pavement performance prediction models using climate, traffic, material, and LTPP data, and implemented budget- and performance-constrained maintenance decision models.
- **Customized Bus and Air–Rail Intermodal Optimization** 2020 – 2022
National University Student Innovation & Entrepreneurship Development Program | Project Leader
 - Led the development of a two-stage optimization framework for air–rail passenger transfers, combining high-speed rail timetable adjustment with customized bus route planning.
 - Modeled the airport–station connection as a vehicle routing problem with time windows and transfers, and developed a hybrid ALNS–Tabu Search algorithm; the resulting software received a national software copyright.
- **Additional Applied Transportation and Infrastructure Research** 2022 – 2024
 - Contributed to research on green transportation and infrastructure lifecycle management, including intelligent inspection, pavement performance prediction, and maintenance decision-making models.
 - Conducted comparative analysis of highway design and acceptance standards for international transportation infrastructure adaptation.

TECHNICAL SKILLS

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- **Programming & Modeling Languages:** Python, MATLAB, C#, C++.
 - **Optimization Solvers & Modeling:** Gurobi, GAMS.
 - **Optimization Methods:** mixed-integer programming, column generation, branch-and-price.
 - **Machine Learning Frameworks:** TensorFlow, PyTorch.
 - **Machine Learning Methods:** graph neural networks, sequence-to-sequence forecasting, transfer learning, multi-task learning.
 - **Transportation & Infrastructure Modeling:** space-time network modeling, path-based traffic assignment, microscopic traffic simulation, travel demand modeling, spatial network analysis, pavement performance prediction, pavement maintenance optimization.
 - **English:** TOEFL 112 (Reading 30, Listening 30, Speaking 25, Writing 27). 01/2025

SELECTED HONORS

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- Second-Class Academic Excellence Scholarship, Tsinghua University 2024
 - Selected to Deliver the Incoming Master's Student Address at Tsinghua University's Graduate Student Opening Ceremony 2022
 - Outstanding Graduate of Beijing Municipality (Top 5%, 19/380) 2022
 - Wu Fu-Zhenhua Transportation Education Excellent Student Award (Top 0.1%, 1/1520) 2021
 - Zhan Tianyou Scholarship (Top 0.2%, 2/1520; Awarded Twice) 2020, 2021
 - Zhixing Scholarship / President's Scholarship (Top 0.1%, 10/16800) 2020